Engineering Mathematics - I · Unit_01 · Unit 1

DRAFT — UE26MA141A · Unit_01 · Unit 1 · Total derivative, Chain rule and Implicit functions

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Total derivative, Chain rule and Implicit functions

1 Total derivative, Chain rule and Implicit functions

Study focus: Total derivative, Chain rule and Implicit functions. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000066 · EV0000067

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Let's welcome you all to another session of partial differentiation.

Key details

- Today we'll be learning chain rule, composite functions, and total derivative of implicit function.
- It's like an extension of what you already did in the case of a function of single variable.
- There are three functions which you need to take it.
- The first one is exponential.
- Next you have to differentiate sine, which will be cos of x square.
- Next you differentiate x square, which will be 2x.
- So that means you had f of g of you had f of g of h of x.
- And when you differentiate it, you followed the order of differentiating f first, then g, and then in the end with respect to h.
- And then if x and y again are functions of one some independent variable, in this case it could be t.
- If you substitute y as 2t square, you will get 2t square, the whole q.
- But always it might not be that easier for me to substitute the independent variables and reduce it to a single variable and differentiate.
- So what we can do is we have another alternate method.
- I had z is equal to x square y cube.
- I took x is equal to t and then I took y is equal to some 2t square.
- What happens if I differentiate x with respect to t, I can get the total derivative because it's a function of single variable.
- I already know the partial derivative of z with respect to x and y.

- Plus dou z by dou y, which will be x square, I am keeping it as it is.
- You differentiate y cube with respect to y.
- T to the power of six.
- I think I took t square if I'm not wrong.

Example or application from the lecture

- For example, suppose I asked you to differentiate this function, which is y is equal to e to the power of sine of x square.
- This was just an example of a function which is a composition of two more functions because it's of single variable, right?
- For example, suppose I have a function z, which is a function of two independent variables x and y.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

Core explanation · 05:00–09:59

You can substitute x square again, so this will be t power 4, 3 into 2 t square, the whole square into 4 t square.

Key details

- And we can simplify this.
- So this is an alternate method of.
- Finding the total derivative of a function of two or more independent variables when each of the independent variables is a function of another variable.
- So that means whenever z is a function of xy.
- And x and y are functions of some parameter say t.
- In that case, I can write.
- I can actually find the total derivative of z with respect to a single variable t.
- I can do that by finding the partial derivative of z with respect to x into the total derivative of x with respect to whatever the variable you have t.
- Plus, I can also do z y dy by d.
- You can apply this formula when the independent variables are functions of a single variable.
- A paraboloid you can imagine like a cup, okay?
- With the base, of course.
- Something like that.
- So a paraboloid and the surface whose Cartesian equation that means I have a cylinder and a paraboloid.
- And then I have a surface whose Cartesian form is cylinder.
- I need to calculate how fast Z is changing when X is equal to two.
- I don't know what is dy by dt is, so I'm not worried about that.
- I need to calculate what is the value of z.
- Do I know z already?
- Yes, z is x square by 16 minus y square by 9.

- with respect to x will be x by eight.
- Partial derivative of z with respect to y will be minus 2y by 9.
- that this is dou z by dou x as z is a function of xy into t x by dt plus.
- Do z by dou y into t y by d.
- This is what I have.
- We can substitute x is equal to 2, that's not a problem.
- I can find this, I already know this, I know this, I know this, but I don't know this.
- So let me erase that first.
- that this is an implicit function, right?
- If y is a function of x, so I can definitely differentiate this, I'll get 2x plus 2y into dy by dx actually equal to 0.

Example or application from the lecture

- So single variable, for example.
- Suppose I have a I have a problem now, okay?
- A point P is moving along the curve of intersection whose Cartesian equation is given.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

Core explanation · 10:00–14:59

Can I do this?

Key details

- I'll get dy by dx equal to minus x by y.
- But I don't want dy by dx.
- What I want is dy by dt.
- These things are known to you when you did the first order partial derivatives.
- Minus x by y.
- You can also find what is the value of y when x is equal to 2.
- when x is equal to 2.
- And what is the formula for dz by dt now?
- You have zx into dx by dt.
- plus ZY into TY by D.
- Zx we got it as x by 8 into dx by dt.
- Z y also we know which is minus 2y by 9.
- dy by dt, I just now got that to be minus x by y into dx by dt which is again 0.2.
- Just put x is equal to 2, you will get the answer.
- If dz by dt is negative you can say that z is decreasing at say x is equal to 2.
- At x is equal to 2, I'll get 0.2 divided by 4, and then plus or minus or plus.
- This will become plus and x is again 2, I have to substitute.
- All I'm interested in to know the behavior of the function.
- They want you to calculate the value of the function at a different point.

- So that means I x is there is a small change in x which is 0.1 and a small change in y which is 0 point.
- Again, you just apply the definition of dz.
- If I don't want to restrict myself to any particular variable, I can write the equation like this also.
- And you substitute the values that are there and you can find that approximately it is equal to 6.418.
- I have another problem.
- If the area of a triangle remains constant,.
- So what is at what rate does the angle between the side should change when the first side is 20 centimeter and the second is 30 centimeter.

Example or application from the lecture

- But first, let's find out what is dy by dt is.
- Suppose now what is required is dz by dt, correct?
- They have already given it as 0.2.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 15:00–19:59

They have given me this side and this side.

Key details

- I want to know at what rate this theta should change, whether theta should increase or theta should decrease, so that the area of the triangle remains constant.
- And the area is half into base into height.
- sine theta will be opposite side by hypotenius.
- That's how I got this expression half x y sin theta.
- So, because there are three independent variables, you calculate the partial derivative with respect to x, dx by dt, with respect to y, dy by dt, with respect to theta, d theta by t.
- And what is required is this, they say that area remains constant.
- So what is the meaning of that area remains constant means dA by dt is 0.
- I don't want any change to happen to this area.
- Again, I have another wherein you have to conclude certain assertions.
- And four feet across the diameter.
- It is known that the steel will expand or contract with temperature changes, right?
- or in the height of the tank.
- You need to tell me whether the change in the diameter or change in height will affect the volume.
- This is also simple.
- And we conclude that dv is equal to 40 pi dr plus 4 pi dh.
- If you observe carefully, so I got an expression that the volume is changing as 40 pi dr plus 4 pi dh.

- Because the coefficient of r is 10, definitely the radius would contribute more towards the change in the volume than the height.
- Just by doing a little bit of differentiation, you can conclude what is the change that is happening, what is contributing to the maximum change.
- And then each of them were function of t.
- This is what I had in the previous case.
- This is what I did.

Example or application from the lecture

- Suppose I'll draw a perpendicular because I need the height.
- Let me take the base as x and suppose this side is y, then that sine theta, right?
- So I hope you appreciate this example so that things will be that much clearer to you.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 20:00–24:59

Let's change little bit.

Key details

- Z is a function of two independent variables, say x as well as y.
- But both x and y are functions of again two independent variables u and v.
- That means what I have to differentiate Z partially.
- But Z is a function of XY.
- Z u will be Zx into XU plus.
- Z y or ZY into Y U.
- I need to calculate this Z U, so I'll calculate Z with respect to X and then with respect to Y and then I'll go like this.
- It can be a function of n variables.
- Then you can find the partial derivative of z with respect to all these three independent variables.
- But you don't need to restrict to only two independent variables, you can move on and find out what is it for a multivariable function as well.
- Then I need to prove that 6vx plus 4v plus 3vz is 0.
- And each of them R is a function of XY, S is a function of X and Y, and T is also a function of X and Y.
- That means I can calculate what is Vx.
- What will be Vx?
- Vx would be VR Rx V S X V T X.
- And it's easy for you to calculate them.
- S X is just minus 4 and S y is just 3.
- And T we have taken it as 4z minus 2x.
- Let me use an eraser.
- this is ZX.

- How do we get Vx?
- Vx is VR R X.
- Then again V S, there is no X there, but you have to write.
- And then V T X.
- So this term here is 0 because I don't have s as a function of x.

Example or application from the lecture

- Let's take an example now.
- They have given V to be a function of this.
- Let's just move a little bit.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 25:00–29:59

I have another example.

Key details

- It's always better if you write the tree diagram every time you have a function of more than one independent variable, so that you won't do any mistake in calculating the partial derivative.
- R is a function of T and S and theta is also a function of T and S.
- I want Zs.
- I have to keep it as it is because I don't know what the function is.
- See if here what will be Rs, Rs will be just T and R T will be S minus 2T.
- And z theta, I don't know, I have to keep it as it is, as I don't know what the function is.
- But theta s of course I know.
- 2 root of s square plus t square into 2 s.
- I can get rid of this.
- But the problem with this problem is you don't know what is this function.
- I need to retain them as it is in the solution as well.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Total derivative, Chain rule and Implicit functions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Total_derivative_Chain_rule_and_Implicit_functions__d194dd0f27d571b2.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Total derivative, Chain rule and Implicit functions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Total_derivative_Chain_rule_and_Implicit_functions__d194dd0f27d571b2.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Total derivative, Chain rule and Implicit functions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Total_derivative_Chain_rule_and_Implicit_functions__d194dd0f27d571b2.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.
Total derivative, Chain rule and Implicit functions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Total_derivative_Chain_rule_and_Implicit_functions__d194dd0f27d571b2.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.

4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00018	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00019	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0059. Sidecar variants compare...
SRC00020	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00021	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0059. Sidecar variants compare...

C Revision and retrieval practice